

Descriptive performance–efficiency ratios (not selection metrics)

Model	Top-1/ms	Macro-F1/ms	Throughput/MiB	Δ Top-1 vs VOMM	Δ p95 ms vs VOMM	Δ stored bytes
Majority baseline	385.28	70.56	7,931,713	-0.2406	-2.095	N/A
First-order Markov	70.59	32.54	1,573,957	-0.0861	-2.086	N/A
Hard-backoff VOMM	0.38	0.19	4,662	+0.0000	+0.000	+0
Interpolated VOMM	13.00	6.27	332,259	-0.0024	-2.035	N/A
Selected Transformer	1.10	0.63	24,714	+0.0857	-1.287	+2,802

Source: overall_metrics.json; efficiency.json; single_checkpoint_evaluation.json · Limitation: Unavailable storage values remain N/A.